Combinatorics: 10 Hard Problems

Full-Topic Coverage, Jane Street Stage-1 Calibration — Each Question Fuses Two or More Techniques

Real Stage-1 quant screens rarely test one identity in isolation — they test whether you *recognize which two ideas to fuse*, under time pressure, before a trap answer looks correct. These 10 problems span the entire combinatorics toolkit (counting rules, permutations/circular arrangements, stars-and-bars, PIE, derangements, Catalan/ballot theory, recursions, cycle structure, and partitions) but every single one requires *stacking* at least two techniques. H6 and H9 lean into Stage-2/Finals territory in raw difficulty — they're included anyway because fluency with them makes everything at Stage-1 feel easy by comparison. Every option set has exactly one correct numeric answer and three answers built from a *specific, common* error — learn to recognize your own mistake in the wrong options, not just find the right one.

Problem H1. *[Technique: Stars-and-Bars + PIE, two-sided bounds]*

How many integer solutions does $x_1 + x_2 + x_3 + x_4 = 24$ have, subject to $2 \le x_i \le 8$ for every i?

Options: (A) 89 (B) 149 (C) 489 (D) 969

Solution. Shift $y_i = x_i - 2 \ge 0$, so $\sum y_i = 24 - 8 = 16$ with cap $y_i \le 6$. Unrestricted: $\binom{16+3}{3} = \binom{19}{3} = 969$. Overflow event $B_i : y_i \ge 7$. Shift again, $\sum = 16 - 7 = 9$: $\binom{9+3}{3} = \binom{12}{3} = 220$ per B_i, times 4 desks = 880. Pairwise ($y_i, y_j \ge 7$ together): $7 + 7 = 14 \le 16$, so possible. Remaining sum $= 16 - 14 = 2$: $\binom{2+3}{3} = \binom{5}{3} = 10$, times $\binom{4}{2} = 6$ pairs = 60. Triple (≥ 7 three at once): $21 > 16$, impossible — stop here. PIE: bad $= 880 - 60 = 820$. Valid $= 969 - 820 = \boxed{149}$.

Trap analysis: 969 = forgot the upper bound entirely. 489 = shifted the overflow threshold on the wrong variable ($x_i \ge 9$ instead of $y_i \ge 7$). 89 = did the single-overflow subtraction but forgot to add back the pairwise term (classic PIE sign slip).

Correct: (B) 149

Problem H2. *[Technique: Circular non-adjacent selection + subcombination]*

20 traders sit around a circular table. First choose 4 non-adjacent seats as "observers" (no two chosen seats adjacent); then choose 2 of those 4 observers to co-sign a report. How many total outcomes?

Options: (A) 2275 (B) 10920 (C) 13650 (D) 14280

Solution. Non-adjacent selection on a circle of n, choosing k: $\frac{n}{n-k}\binom{n-k}{k}$. With $n = 20, k = 4$: $\binom{16}{4} = 1820$, and $\frac{20}{16} \times 1820 = 2275$ ways to pick the observers. Then choose 2 of the 4 for the joint report: $\binom{4}{2} = 6$. Total: $2275 \times 6 = \boxed{13650}$.

Trap analysis: 14280 = used the *linear-row* non-adjacent formula $\binom{n-k+1}{k} = \binom{17}{4} = 2380$ instead of the circular one — the single most common error on circular selection problems. 10920 = used $\binom{16}{4}$ alone, forgetting the $\frac{n}{n-k}$ circular correction factor. 2275 = correct observer count, but forgot the second (sub-committee) stage entirely.

Correct: (C) 13650

▶ *Speed insight:* On a circle, "choose k non-adjacent" is never just $\binom{n-k}{k}$ — that undercounts because it doesn't distinguish which "gap" wraps around seat $n \to 1$. The $\frac{n}{n-k}$ factor fixes exactly that.

Problem H3. *[Technique: Fixed points via $\binom{n}{k} D_{n-k}$]*

7 analysts each get a random report back (uniform over all 7! assignments). In how many of these assignments do *exactly* 3 analysts get their own correct report back (and the other 4 are fully deranged)?

Options: (A) 70 (B) 315 (C) 840 (D) 5040

Solution. Choose which 3 are correct: $\binom{7}{3} = 35$. The remaining 4 must be deranged (zero fixed points among themselves): $D_4 = 9$.

$$35 \times 9 = \boxed{315}.$$

Sanity check: $P(\text{exactly 3 correct}) = 315/5040 = 1/16$ exactly — a suspiciously clean fraction, which is a good sign you didn't make an arithmetic slip.

Trap analysis: $840 = 35 \times 4!$, i.e. forgot to derange the remaining 4 and instead let them permute freely (double-counts the cases where some of those 4 also land correctly). $70 = \binom{7}{4} \times D_3 = 35 \times 2$, i.e. accidentally swapped which group (3 vs. 4) gets fixed vs. deranged. 5040 = forgot to restrict at all.

Correct: (B) 315

Problem H4. *[Technique: PIE with a divisibility trap (coincident LCMs)]*

How many four-digit integers (from 1000 to 9999) are divisible by 6, 10, or 15?

Options: (A) 1500 (B) 2100 (C) 2400 (D) 3000

Solution. Here's the trap: $\text{lcm}(6, 10) = \text{lcm}(6, 15) = \text{lcm}(10, 15) = \text{lcm}(6, 10, 15) = 30$ — *all* pairwise and the triple LCM coincide. Counts in $[1000, 9999]$: div. by 6: $1666 - 166 = 1500$; by 10: $999 - 99 = 900$; by 15: $666 - 66 = 600$; by 30: $333 - 33 = 300$.

$$\text{Singles} = 1500 + 900 + 600 = 3000.$$

$$\text{Pairs (each} = 300) : 3 \times 300 = 900. \qquad \text{Triple} = 300.$$

$$|A \cup B \cup C| = 3000 - 900 + 300 = \boxed{2400}.$$

Trap analysis: 3000 = added the three singles and stopped, i.e. forgot PIE corrections entirely (badly overcounts numbers divisible by more than one). 2100 = $3000 - 900$ = subtracted the pairwise term but forgot to add the triple back. 1500 = only counted "divisible by 6" and ignored the other two sets.

Correct: (C) 2400

▶ *Speed insight:* When several pairwise LCMs come out equal, always double-check whether the *triple* LCM also matches — if all four PIE terms collapse to the same divisor, you can factor the whole computation as (3 singles) $- 2 \times$ (count for that one divisor), which is much faster than writing out six separate numbers.

Problem H5. *[Technique: Composition with a two-sided cap — the "deficit" symmetry trick]*

In how many ways can 15 be written as an ordered sum of 4 positive integers, each between 1 and 5 inclusive?

Options: (A) 28 (B) 52 (C) 56 (D) 364

Solution. *Direct method.* Shift $y_i = t_i - 1 \in [0, 4]$, target $\sum y_i = 15 - 4 = 11$. Unrestricted: $\binom{11+3}{3} = \binom{14}{3} = 364$. Overflow $y_i \ge 5$: shifted sum $= 11 - 5 = 6$, $\binom{6+3}{3} = \binom{9}{3} = 84$, times 4 = 336. Pairwise ($y_i, y_j \ge 5$): $5 + 5 = 10 \le 11$, shifted sum = 1: $\binom{1+3}{3} = 4$, times $\binom{4}{2} = 6$: 24. Triple: $15 > 11$, impossible. Bad $= 336 - 24 = 312$. Valid $= 364 - 312 = \boxed{52}$.

Symmetry shortcut. Since 15 is close to the maximum possible sum $4 \times 5 = 20$, work with the *deficit* $d_i = 5 - t_i \in [0, 4]$ instead: $\sum d_i = 20 - 15 = 5$. This is now a *small*, uncapped-in-practice stars-and-bars count (max possible $d_i = 4$, target only 5, so at most one d_i can hit the cap): $\binom{5+3}{3} = \binom{8}{3} = 56$, minus the single case where some $d_i \ge 5$ (shifted sum = 0, giving exactly 1 way, times 4 desks = 4): $56 - 4 = 52$. Same answer, far less arithmetic.

Trap analysis: 364 = forgot the upper cap entirely (plain lower-bound stars-and-bars). 56 = used the deficit trick's *unrestricted*

count without the small overflow correction. 28 = computed the direct method but forgot to add back the pairwise term.

Correct: (B) 52

▶ *Speed insight:* Whenever a stars-and-bars target is close to the maximum allowed sum (rather than close to the minimum), flip to counting the "deficit from max" instead — the overflow corrections shrink dramatically and often vanish after just one PIE term.

Problem H6. *[Technique: Generalized ballot theorem (asymmetric Dyck path)]*

A book executes 12 trades: 8 buys (+1) and 4 sells (−1), in some order (net position ends at +4). In how many orderings does the running inventory stay ≥ 0 the entire time?

Options: (A) 220 (B) 275 (C) 429 (D) 495

Solution. Total orderings: $\binom{12}{4} = 495$. Reflect the prefix of any "bad" path (one that first touches -1) across the line $y = -1$: this bijects bad paths onto *all* 8-up/4-down paths ending at $-2-(8-4) = -6$, which forces the reflected path to have $q-1 = 3$ ups and $p+1 = 9$ downs, i.e. $\binom{12}{3} = 220$ bad paths.

$$\text{Good} = 495 - 220 = \boxed{275}.$$

This matches the general ballot/cycle-lemma formula directly: for p ups, q downs ($p \geq q$), the count of paths staying ≥ 0 is $\frac{p-q+1}{p+1}\binom{p+q}{q} = \frac{5}{9}(495) = 275$.

Trap analysis: 495 = forgot to exclude any bad paths at all. 220 = reported the *bad*-path count instead of subtracting it from the total. 429 (which is actually C_7, a real Catalan number!) is the trap for anyone who pattern-matches "Dyck path" and reflexively reaches for the equal-steps Catalan formula without noticing $p \neq q$ here.

Correct: (B) 275

Problem H7. *[Technique: Recursive/bijective counting — domino tilings]*

Let $f(n)$ be the number of ways to tile a $2 \times n$ strip using only 1×2 dominoes (horizontal or vertical). Set up the recurrence for $f(n)$ and compute $f(10)$.

Options: (A) 34 (B) 55 (C) 89 (D) 144

Solution. Condition on how column 1 is covered: either one *vertical* domino fills it entirely, leaving a $2 \times (n-1)$ strip to tile in $f(n-1)$ ways; or two *horizontal* dominoes stacked span columns 1–2, leaving a $2 \times (n-2)$ strip in $f(n-2)$ ways. These two cases are exhaustive and disjoint, so

$$f(n) = f(n-1) + f(n-2), \qquad f(0) = 1,\ f(1) = 1.$$

This is the Fibonacci recurrence (shifted): $f(n) = F_{n+1}$. Iterating: $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89$ for $n = 0, \ldots, 10$.

$$f(10) = \boxed{89}.$$

Trap analysis: 55 = $f(9)$ and 144 = $f(11)$ are both off-by-one indexing errors — the single most common mistake with Fibonacci-type recurrences is misaligning which term the initial conditions $f(0), f(1)$ actually correspond to. 34 = $f(8)$, off by two.

Correct: (C) 89

▶ *Speed insight:* First-step (or first-column) decomposition is the master technique behind *every* linear-recurrence combinatorics question: identify the finitely many disjoint, exhaustive things that can happen to the very first unit, and the recurrence falls out immediately — no generating functions needed at Stage-1 speed.

Problem H8. *[Technique: Cycle-type counting — derangements aren't enough]*

In how many permutations of $\{1, \ldots, 6\}$ do *all* cycles have length at least 3 (i.e., no fixed points *and* no 2-cycles)?

Options: (A) 40 (B) 120 (C) 160 (D) 265

Solution. A permutation's cycle type is a partition of $n = 6$; we need only parts ≥ 3: the partitions of 6 into parts ≥ 3 are (6) and (3, 3). The number of permutations with cycle type $1^{a_1}2^{a_2}\cdots n^{a_n}$ is $\frac{n!}{\prod_i i^{a_i} a_i!}$ (divide the $n!$ orderings by the rotations within each cycle and the swaps between equal-length cycles). Type (6): $\frac{6!}{6^1 \cdot 1!} = \frac{720}{6} = 120$ (equivalently, the number of n-cycles is always $(n-1)!$). Type (3, 3): $\frac{6!}{3^2 \cdot 2!} = \frac{720}{18} = 40$.

$$\text{Total} = 120 + 40 = \boxed{160}.$$

Trap analysis: 265 = D_6, the plain derangement count — this is the single most dangerous trap on the whole sheet, because it's the "obviously relevant" number that answers a subtly different question (excludes only fixed points, not 2-cycles). 120 = counted only the 6-cycles. 40 = counted only the (3, 3)-type.

Correct: (C) 160

▶ *Speed insight:* The instant a problem forbids *more* than just fixed points (e.g. "no fixed points and no 2-cycles"), plain derangements no longer apply — you need the cycle-type/cycle-index formula. Recognizing this distinction under time pressure is exactly what separates Stage-1 survivors from the rest.

Problem H9. *[Technique: Catalan convolution — first return to zero]*

A Dyck path has 10 up-steps and 10 down-steps, starting and ending at 0, staying ≥ 0 throughout (one of the C_{10} such paths). In how many of them does the walk return to 0 for the *first* time (after the start) exactly at step 6?

Options: (A) 264 (B) 858 (C) 2145 (D) 16796

Solution. If a semilength-n Dyck path first returns to 0 at step $2k$, the middle $2k-2$ steps (strictly between the first up-step and the matching down-step back to 0) form an independent, "elevated" Dyck path of semilength $k-1$, and the remaining $2(n-k)$ steps after the first return form an independent Dyck path of semilength $n-k$. This gives the Catalan convolution:

$$\text{count} = C_{k-1} \cdot C_{n-k}.$$

Here $2k = 6 \Rightarrow k = 3$, $n = 10$: count $= C_2 \cdot C_7 = 2 \times 429 = \boxed{858}$.

Trap analysis: 2145 = $C_3 \times C_7$: forgot the "-1" and used k instead of $k-1$. 264 = $C_2 \times C_6$: used $n-k-1$ instead of $n-k$ for the second factor. 16796 = C_{10}: gave the count of *all* Dyck paths, ignoring the "first return at step 6" condition entirely.

Correct: (B) 858

Problem H10. *[Technique: Euler's partition theorem — odd parts vs. distinct parts]*

Let $p_{\text{odd}}(12)$ be the number of partitions of 12 into parts that are all odd (repeats allowed), and $p_{\text{distinct}}(12)$ the number of partitions of 12 into parts that are all distinct. Without enumerating $p_{\text{odd}}(12)$ directly, find the common value $p_{\text{odd}}(12) = p_{\text{distinct}}(12)$.

Options: (A) 12 (B) 15 (C) 18 (D) 77

Solution. *Euler's theorem* guarantees $p_{\text{odd}}(n) = p_{\text{distinct}}(n)$ for every n, via Glaisher's bijection: given a partition into odd parts, group equal parts (say odd value m occurs c_m times), write c_m in binary as a sum of distinct powers of 2, and split those c_m copies of m into the parts $m \cdot 2^{a_1}, m \cdot 2^{a_2}, \ldots$. Because m is odd, each resulting part factors uniquely as (odd)×(power of 2), so all parts produced this way — across every distinct m — are themselves automatically distinct. The map is reversible (recover m and the

powers of 2 from each part's factorization, then re-sum the binary digits), so it's a genuine bijection.

This means we only need to count the (much easier to enumerate) distinct-parts partitions of 12, organized by largest part:

$$\{12\};\ \{11,1\};\ \{10,2\};\ \{9,3\},\{9,2,1\};\ \{8,4\},\{8,3,1\};$$

$$\{7,5\},\{7,4,1\},\{7,3,2\};\ \{6,5,1\},\{6,4,2\},\{6,3,2,1\};\ \{5,4,3\},\{5,4,2,1\}.$$

Counting: $1+1+1+2+2+3+3+2=\boxed{15}$.

Trap analysis: $77 = p(12)$, the *total* (unrestricted) partition count of 12 — forgetting the odd/distinct restriction is the obvious trap. 12 and 18 are $p_{\text{distinct}}(11)$ and $p_{\text{distinct}}(13)$ respectively — off-by-one errors from misindexing n.

Correct: (B) 15

▶ *Speed insight:* Whenever a problem hands you two combinatorial objects and asks you to show they're equinumerous, look for a classical bijection theorem first (Euler odd↔distinct, or the Catalan-family bijections from H6/H9) before attempting to count either side from scratch — it's almost always faster to count the easier side once you've justified the bijection.

Calibration note: H1–H5 and H8 are realistic 90–150s Stage-1 questions once the fusion is recognized. H6 and H9 (ballot theorem, Catalan convolution) and H10 (Glaisher's bijection) are genuinely Stage-2/Finals-grade — include them in your drilling anyway, because half of getting Stage-1 speed is having already seen the "scary" version of an idea so the Stage-1 version looks trivial by comparison.